Homework 1

BSTA 551

Due 01/15 at 11pm

Directions

Please turn in this homework on Sakai. Please submit your homework in pdf format or in html rendered from a qmd (or Rmd) file (please also include source qmd or Rmd). You can type your work on your computer or submit a single file with photos of your written work or any other method that can be turned into a pdf.

Try to complete all of the problems listed below at some point this quarter! You may want to save some of them for studying later!

You must show all of your work to receive credit.

This homework covers material from Lessons 1-3 (Week 1): sampling distributions, point estimation, bias, variance, MSE, and minimum variance unbiased estimation.

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this github link

Questions

Problem 1: Sampling Distribution of the Sample Mean

Researchers are studying the birth weights of full-term infants at a large hospital. Let $X_1, X_2, \ldots, X_n$ be a random sample of birth weights from a population with mean $\mu = 3400$ grams and standard deviation $\sigma = 450$ grams.

a. For a sample of size $n = 25$, calculate:

   - $E(\bar{X})$

- $\text{Var}(\bar{X})$
- $SE(\bar{X})$ (the standard error of the sample mean)

b. How large must the sample size n be so that the standard error of $\bar{X}$ is at most 50 grams?

c. Suppose we define a new statistic $W = \frac{1}{n}\sum_{i=1}^{n}(X_i - 3000)$. Find $E(W)$ and $\text{Var}(W)$ in terms of μ, σ^2, and n. Is W an unbiased estimator of $\mu - 3000$?

Problem 2: Bias, Variance, and MSE

Let $X_1, X_2, \dots, X_n$ be a random sample from an Exponential distribution with rate parameter $\lambda > 0$ (so that $E(X_i) = 1/\lambda$ and $\text{Var}(X_i) = 1/\lambda^2$).

a. Consider the estimator $\hat{\mu} = \bar{X}$ for the population mean $\mu = 1/\lambda$.

- Show that $\hat{\mu}$ is an unbiased estimator of μ.
- Calculate $\text{Var}(\hat{\mu})$ in terms of λ and n.

b. Consider the estimator $\hat{\lambda} = 1/\bar{X}$ for the rate parameter λ.

- It can be shown that $E(1/\bar{X}) = \frac{n\lambda}{n-1}$ for $n > 1$. Calculate the bias of $\hat{\lambda}$.
- Propose a modification to make $\hat{\lambda}$ unbiased. Call this estimator $\tilde{\lambda}$.

c. For $n = 12$ and $\lambda = 0.5$:

- Calculate the bias of $\hat{\lambda} = 1/\bar{X}$
- Given that $\text{Var}(1/\bar{X}) = \frac{n^2\lambda^2}{(n-1)^2(n-2)}$ for $n > 2$, calculate the MSE of $\hat{\lambda}$

d. Write R code to verify your answers from part (c) using simulation with 10,000 replications.

Problem 3: The Bias-Variance Tradeoff

A hospital infection control team wants to estimate the proportion p of patients who develop hospital-acquired infections during their stay. Out of $n = 40$ patients, X patients develop an infection where $X \sim \text{Binomial}(n, p)$.

Two estimators are proposed:

- Standard estimator: $\hat{p}_1 = \frac{X}{n}$
- "Add-one" estimator: $\hat{p}_2 = \frac{X+1}{n+2}$

a. Show that $\hat{p}_1$ is unbiased for p. What is its variance?

b. Calculate the expected value of $\hat{p}_2$ and find its bias. Show that the bias shrinks the estimate toward 0.5.

c. For $n = 40$ and $p = 0.05$ (a low infection rate), calculate:

- MSE of $\hat{p}_1$
- Variance of $\hat{p}_2$ is $\frac{np(1-p)}{(n+2)^2}$. Calculate the MSE of $\hat{p}_2$

Which estimator is preferred based on MSE?

d. Using R, create a plot comparing the MSE of $\hat{p}_1$ and $\hat{p}_2$ across all values of p from 0.01 to 0.99 for $n = 40$. At what range of p values does the add-one estimator outperform the standard estimator?

Problem 4: Sample Variance Estimators

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Consider two estimators for σ^2:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \quad \text{and} \quad \tilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

a. It is known that $E(S^2) = \sigma^2$ (i.e., S^2 is unbiased). Use this fact to find $E(\tilde{S}^2)$ and calculate the bias of $\tilde{S}^2$.

b. For $n = 8$ and $\sigma^2 = 64$, what is the bias of $\tilde{S}^2$? What percentage of the true variance does this bias represent?

c. Use R to simulate 10,000 samples of size $n = 8$ from a $N(0, 8)$ distribution (so $\sigma^2 = 64$). Calculate S^2 and $\tilde{S}^2$ for each sample and verify that:

- The mean of S^2 values is approximately 64
- The mean of $\tilde{S}^2$ values matches your theoretical result from part (a)

Problem 5: Estimating Vaccine Efficacy

In a vaccine trial, researchers want to estimate the efficacy rate p (proportion of vaccinated individuals who develop immunity).

a. In a pilot study, $n = 60$ vaccinated individuals are tested, and $X = 51$ show evidence of immunity. Calculate:

- The point estimate $\hat{p}$
- The estimated standard error of $\hat{p}$

Hint: Use $\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

b. If the true efficacy is $p = 0.80$, calculate:

- The theoretical MSE of $\hat{p}$
- The theoretical standard error of $\hat{p}$

c. For the full trial, the researchers want the standard error of their efficacy estimate to be at most 0.04. What is the minimum sample size required? Assume $p \approx 0.85$ for planning purposes.

Problem 6: Devore et al. Section. 7.1 Exercise 22

Return to the problem of estimating the population proportion p and consider another adjusted estimator, namely

$$\hat{P} = \frac{X + \sqrt{n/4}}{n + \sqrt{n}}$$

where $X \sim \text{Binomial}(n, p)$.

(The justification for this estimator comes from the Bayesian approach to point estimation.)

a. Determine the mean squared error of this estimator. What is interesting about this MSE?

b. Compare the MSE of this estimator to the MSE of the usual estimator (the sample proportion $\hat{p} = X/n$).